Learning a function energy space

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1 Preliminaries

Definition 1 (Polish space). *A metric space (X, d) is called a* Polish space *if it is*

1. *complete, i.e., every Cauchy sequence in (X, d) converges to a point in X;*
2. *separable, i.e., there exists a countable dense subset $D \subset X$.*

Typical examples of Polish spaces include $\mathbb{R}^n$ with the Euclidean metric, separable Banach spaces, Hilbert spaces, and function spaces such as $L^2(M)$ or $H^s(M)$ when M is a compact manifold.

Definition 2 (Sigma-algebra). *Let X be a set. A collection $\mathcal{A} \subset 2^X$ is called a* σ-algebra *on X if:*

1. *$X \in \mathcal{A}$;*
2. *if $A \in \mathcal{A}$, then $A^c := X \setminus A \in \mathcal{A}$;*
3. *if $\{A_n\}_{n=1}^{\infty} \subset \mathcal{A}$, then*
$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}.$$

It follows immediately that $\varnothing \in \mathcal{A}$ and that $\mathcal{A}$ is also closed under countable intersections.

Definition 3 (Borel σ-algebra). *Let (X, d) be a metric space. The* Borel σ-algebra *on X, denoted by $\mathcal{B}(X)$, is the smallest σ-algebra containing all open subsets of X:*

$$\mathcal{B}(X) := \sigma\big(\{U \subset X : U \text{ is open}\}\big).$$

If X is Polish, then $(X, \mathcal{B}(X))$ enjoys strong regularity properties that make it the canonical measurable structure for probability theory.

Definition 4 (Measurable function). *Let $(X, \mathcal{B}(X))$ be a measurable space. A function*

$$R : X \to [0, \infty]$$

is said to be measurable *if for every Borel set $A \subset [0, \infty]$,*

$$R^{-1}(A) \in \mathcal{B}(X).$$

Equivalently, R is measurable if and only if the sublevel sets

$$\{f \in X : R(f) \le t\} \in \mathcal{B}(X) \quad \text{for all } t \in \mathbb{R}.$$

Definition 5 (Gibbs measure induced by a regularizer). *Let (X, d) be a Polish space, let $\mathcal{B}(X)$ be its Borel σ-algebra, and let ν be a σ-finite Borel measure on X. Let $R : X \to [0, \infty]$ be a measurable functional such that for some $\lambda > 0$,*

$$Z_\lambda := \int_X e^{-\lambda R(f)}\, \nu(df) < \infty.$$

The Gibbs measure *associated with R and ν is the probability measure π_λ on $(X, \mathcal{B}(X))$ defined by*

$$\pi_\lambda(df) = \frac{1}{Z_\lambda} e^{-\lambda R(f)}\, \nu(df).$$

Equivalently, π_λ is absolutely continuous with respect to ν, with Radon–Nikodym derivative

$$\frac{d\pi_\lambda}{d\nu}(f) = \frac{1}{Z_\lambda} e^{-\lambda R(f)}.$$

Given observations y with likelihood $p(y \mid f)$, the posterior distribution induced by the Gibbs prior is

$$\pi_\lambda(df \mid y) \propto p(y \mid f)\, e^{-\lambda R(f)}\, \nu(df).$$

The maximum a posteriori (MAP) estimator is therefore given by

$$\widehat{f}_{\text{MAP}} = \arg\min_{f \in X} \Big(-\log p(y \mid f) + \lambda R(f) \Big),$$

which coincides with regularized empirical risk minimization.

2 Learning the regularizer

Assume, the regularizer depends on some parameter θ. Then learning it equivalent to minimizing the negative log-likelihood:

$$\min_\theta \mathcal{L}(\theta) = \lambda \int_f R_\theta(f) d\pi(f) + \log Z(\theta)$$

where $Z(\theta) = \int_{f \in X} \exp(-R_\theta(f))\nu(df)$ is a normalizing constant, X is function space, ν base reference measure on X.

Or, if we are given the empirical observations of functions $\{f_i\}_{i=1}^t$ then

$$\min_\theta \mathcal{L}(\theta) = \frac{\lambda}{t} \sum_i R_\theta(f_i) + \log Z(\theta)$$

Assuming $R_\theta(f) = \langle \theta, \Phi(f) \rangle$ (linear in θ) out problem is convex, yaaaaay! Parameter θ can be here an infinite-dimensional measure or a vector, hence, the minimization problem can be solved by gradient descent on θ.

The gradient of the data term is

$$\nabla_\theta \left(\frac{1}{T} \sum_{t=1}^{T} \langle \theta, \Phi(f^{(t)}) \rangle \right) = \frac{1}{T} \sum_{t=1}^{T} \Phi(f^{(t)}).$$

For the partition function term, differentiation under the integral yields

$$\begin{aligned}
\nabla_\theta \log Z(\theta) &= \frac{1}{Z(\theta)} \nabla_\theta Z(\theta) \\
&= \frac{1}{Z(\theta)} \int (-\Phi(f)) \exp\big(-\langle \theta, \Phi(f) \rangle \big)\, \nu(df) \\
&= -\mathbb{E}_{f \sim p_\theta}[\Phi(f)]\,.
\end{aligned}$$

Combining the two terms, the gradient of the objective is

$$\boxed{\nabla_\theta \mathcal{L}(\theta) = \mathbb{E}_{\text{data}}[\Phi(f)] - \mathbb{E}_{f \sim p_\theta}[\Phi(f)]\,,}$$

where

$$\mathbb{E}_{\text{data}}[\Phi(f)] := \frac{1}{T} \sum_{t=1}^{T} \Phi(f^{(t)}).$$

A gradient descent step with step size $\eta > 0$ is therefore given by

$$\boxed{\theta_{k+1} = \theta_k - \eta \left(\mathbb{E}_{\text{data}}[\Phi(f)] - \mathbb{E}_{f \sim p_{\theta_k}}[\Phi(f)] \right).}$$

If θ is constrained (e.g., $\theta \geq 0$ or θ lies in a simplex), the update is followed by projection onto the feasible set Θ:

$$\theta_{k+1} = \Pi_\Theta \left[\theta_k - \eta \left(\mathbb{E}_{\text{data}}[\Phi(f)] - \mathbb{E}_{f \sim p_{\theta_k}}[\Phi(f)] \right) \right].$$

At a stationary point $\theta^\star$, the optimality condition is the moment-matching identity

$$\mathbb{E}_{\text{data}}[\Phi(f)] = \mathbb{E}_{f \sim p_{\theta^\star}}[\Phi(f)]\,.$$

$\Phi(f)$ here a statistic extractor on function space.

3 Examples

Note, in all the examples that follow $\Phi(f)$ is convex in f.

Example 1 (Energy regularization).

$$\Phi : L^2(M) \to \mathbb{R}, \qquad \Phi(f) := \|f\|^2_{L^2(M)}.$$

The induced regularizer $R(f) = \|f\|^2_{L^2}$ corresponds to ridge/Tikhonov regularization. This choice measures only the magnitude of f and ignores geometric structure.

3.1 First-Order Smoothness

Example 2 (Sobolev H^1 regularization).

$$\Phi : H^1(M) \to \mathbb{R}, \qquad \Phi(f) := \int_M \|\nabla f(x)\|^2 \, dx.$$

This feature measures global smoothness. The null space consists of constant functions.

Example 3 (Anisotropic Sobolev regularization). Let $\{Q_j\}_{j=1}^k$ be fixed symmetric positive semidefinite matrices. Define

$$\Phi : H^1(M) \to \mathbb{R}^k, \qquad \Phi_j(f) := \int_M \langle \nabla f(x), Q_j \nabla f(x) \rangle \, dx.$$

Then

$$R_\theta(f) = \sum_{j=1}^k \theta_j \Phi_j(f)$$

corresponds to learning a Riemannian metric.

3.2 Total Variation and Sparsity

Example 4 (First-order total variation).

$$\Phi : BV(M) \to \mathbb{R}_+, \qquad \Phi(f) := \|\nabla f\|_{\mathcal{M}}.$$

This feature measures the total mass of the gradient as a Radon measure and promotes piecewise-constant functions.

3.3 Second-Order and Curvature-Based Statistics

Example 5 (Second-order total variation / splines).

$$\Phi : X \to \mathbb{R}_+, \qquad \Phi(f) := \|\Delta f\|_{\mathcal{M}},$$

where X is the space of functions with measure-valued Laplacian.

The null space consists of affine functions, and minimizers are adaptive splines.

Example 6 (Anisotropic second-order TV). Let $\mathcal{S}_+$ denote the cone of symmetric positive semidefinite matrices. Define

$$\Phi : X \to \mathcal{C}(\mathcal{S}_+, \mathbb{R}_+), \qquad \Phi(f)(Q) := \|\Delta_Q f\|_{\mathcal{M}},$$

where $\Delta_Q f := \nabla \cdot (Q \nabla f)$.

Here:

- domain: functions with measure-valued anisotropic Laplacian,
- codomain: nonnegative functions on $\mathcal{S}_+$,
- regularizer:

$$R_\theta(f) = \int_{\mathcal{S}_+} \|\Delta_Q f\|_{\mathcal{M}} \, d\theta(Q).$$

This promotes sparse, geometry-aligned curvature and induces a Finsler total variation.

3.4 Spectral Statistics

Example 7 (Spectral energy features). Let $\{\psi_k\}$ be eigenfunctions of a Laplacian on M. Define

$$\Phi : L^2(M) \to \ell^1, \qquad \Phi_k(f) := |\langle f, \psi_k \rangle|^2.$$

The corresponding regularizer weights energy by frequency and induces a spectral bias.

3.5 Measure-Based and Neural Representations

Example 8 (Barron / ReLU representation). Let

$$f_\mu(x) = \int_{\mathbb{S}^{d-1}} \sigma(\langle w, x\rangle)\, d\mu(w),$$

with μ a finite signed Radon measure. Define

$$\Phi : \mathcal{B} \to \mathcal{M}_+(\mathbb{S}^{d-1}), \qquad \Phi(f_\mu) := |\mu|.$$

Here:

- domain: Barron space $\mathcal{B}$,
- codomain: finite nonnegative measures,
- regularizer: $R(f_\mu) = \|\mu\|_{\mathrm{TV}}$.

This promotes sparse ridge representations.

Example 9 (Finsler–TV / Radon–Laplacian). Combining the previous examples,

$$\Phi : \mathcal{B} \to \mathcal{C}(\mathcal{S}_+, \mathbb{R}_+), \qquad \Phi(f_\mu)(Q) := \int \sqrt{w^\top Q w}\, |\mu|(dw).$$

Equivalently,

$$R(f_\mu) = \int_{\mathcal{S}_+} \Phi(f_\mu)(Q)\, d\theta(Q) = \int F(w)\, |\mu|(dw), \quad F(w) := \int \sqrt{w^\top Q w}\, d\theta(Q).$$

This is a Finsler total variation norm on kink directions.

4 Back to the problem

Given all distribution over functions minimizing the negative log-likelihood is

$$\min_\theta \mathcal{L}(\theta) = \lambda \int_f R_\theta(f) d\pi(f) + \log Z(\theta)$$

where $Z(\theta) = \int_{f \in X} \exp(-R_\theta(f))\nu(df)$ is a normalizing constant, X is function space, ν base reference measure on X.

However, everything we observe is just a discrete sample. Hence, given the empirical observations of functions $\{f_i\}_{i=1}^t$ our minimization looks like

$$\min_\theta \mathcal{L}(\theta) = \frac{\lambda}{t} \sum_i R_\theta(f_i) + \log Z(\theta)$$

Again, **everything** we observe is a discrete sample. Usually, we have access to each function f via its values: $D = \{(x_i, y_i)\}$ where $y_i = f(x_i) + \mathcal{N}(0, \sigma^2)$. Note, that

$$\log p_\theta(D_Y | f, D_X) \propto -\frac{1}{\sigma^2} \sum_{(x,y)\in D} \|y - f(x)\|_2^2 = \ell(D, f)$$

Hence,

$$-\log p_\theta\left(D_Y \mid D_X\right) = -\log \int \exp\left(-\ell(D, f) - \lambda R_\theta(f)\right) \nu(df) + \log Z(\theta)$$

So given datasets $\{D^{(i)}\}_{i=1}^t$ our optimization problem is

$$\min_\theta \mathcal{L}(\theta) = -\frac{1}{t} \sum_{i=1}^t \log \int \exp\left(-\ell\left(D^{(i)}, f\right) - \lambda R_\theta(f)\right) \nu(df) + \log Z(\theta) \tag{1}$$

However, the term $\log \int \exp \ldots$ is intractable :(This is why the masking comes into play in practice.

5 Masking

5.1 Posterior concentration around the MAP estimator

Fix a parameter θ and a dataset $D = \{(x_j, y_j)\}_{j=1}^n$. Define the energy functional

$$E_\theta(f) := \ell(D, f) + \lambda R_\theta(f),$$

where $\ell(D, f)$ denotes the empirical negative log-likelihood and R_θ is a regularizer.

Assume the following:

1. The dataset D is large (large n) and noise variance σ^2 is small
2. (*Coercivity*) The functional R_θ is coercive on X modulo a finite-dimensional nullspace $\mathcal{N}$, i.e., $\|f\|_X \to \infty$ with $f \perp \mathcal{N}$ implies $R_\theta(f) \to \infty$.
3. (*Strict convexity*) The functional $\ell(D, \cdot)$ is strictly convex on $X/\mathcal{N}$.
4. (*Regularity*) The map $f \mapsto E_\theta(f)$ is twice differentiable in a neighborhood of its minimizer.

Under these assumptions, the minimization problem

$$\widehat{f}_\theta \in \arg\min_{f \in X} E_\theta(f)$$

admits a unique solution modulo $\mathcal{N}$. Moreover, the posterior distribution

$$p_\theta(f \mid D) \propto \exp\big(-E_\theta(f)\big)$$

concentrates in probability around $\widehat{f}_\theta$ as either $n \to \infty$ or $\sigma^2 \to 0$. In particular, for any neighborhood U of $\widehat{f}_\theta$ in X,

$$p_\theta(U \mid D) \longrightarrow 1.$$

5.2 Laplace approximation of the marginal likelihood

Let $E_\theta(f) = \ell(D, f) + \lambda R_\theta(f)$ and let $\widehat{f}_\theta$ denote its unique minimizer. Define the Hessian operator

$$H_\theta := \nabla^2 E_\theta(\widehat{f}_\theta),$$

acting on the tangent space of $X/\mathcal{N}$.

By a second-order Taylor expansion,

$$E_\theta(f) = E_\theta(\widehat{f}_\theta) + \frac{1}{2}\langle f - \widehat{f}_\theta,\, H_\theta(f - \widehat{f}_\theta)\rangle + o(\|f - \widehat{f}_\theta\|_X^2).$$

Substituting this expansion into the marginal likelihood integral yields the Laplace approximation

$$\int_X \exp\big(-E_\theta(f)\big)\, \nu(df) \;\approx\; \exp\big(-E_\theta(\widehat{f}_\theta)\big) \int \exp\big(-\tfrac{1}{2}\langle h, H_\theta h\rangle\big)\, dh. \tag{2}$$

The remaining integral is Gaussian and evaluates (up to a multiplicative constant) to $(\det H_\theta)^{-1/2}$, where $\det H_\theta$ denotes a suitable regularized determinant. Consequently,

$$\int_X \exp\big(-\ell(D, f) - \lambda R_\theta(f)\big)\, \nu(df) \;\approx\; \exp\big(-\ell(D, \widehat{f}_\theta) - \lambda R_\theta(\widehat{f}_\theta)\big)(\det H_\theta)^{-1/2}. \tag{3}$$

5.3 Decomposition of the marginal likelihood objective

Substituting the Laplace approximation (3) into the negative log marginal likelihood yields

$$-\log p_\theta(D_Y \mid D_X) \;\approx\; \ell(D, \widehat{f}_\theta) + \lambda R_\theta(\widehat{f}_\theta) + \frac{1}{2}\log\det H_\theta - \log Z(\theta), \tag{4}$$

up to additive constants independent of θ.

This expression decomposes the marginal likelihood into four interpretable components:

1. *Data-fit term* $\ell(D, \widehat{f}_\theta)$, measuring how well the MAP estimator explains the observations;
2. *Regularization cost* $\lambda R_\theta(\widehat{f}_\theta)$, encoding alignment between the learned function and the inductive bias imposed by θ;
3. *Local complexity (Occam) penalty* $\frac{1}{2}\log\det H_\theta$, which penalizes flat or unstable directions around the MAP solution;
4. *Global complexity penalty* $-\log Z(\theta)$, measuring the overall volume of functions favored by the prior.

Together, these terms characterize the tradeoff between data fit, regularization, and model complexity in the marginal likelihood objective.

6 Back to the optimization

Given the collection of datasets $\mathcal{D} = \left\{ \left(D_{\text{train}}^{(i)}, D_{\text{test}}^{(i)} \right) \right\}_{i=1}^{t}$ Substituting this into Equation (1) we have

$$\min_{\theta} \mathcal{L}(\theta) = \frac{1}{t} \sum_{i=1}^{t} \ell \left(D_{\text{test}}^{(i)}, \widehat{f}_{\theta}^{(i)} \right) + \lambda R_{\theta} \left(\widehat{f}_{\theta}^{(i)} \right) + \frac{1}{2} \log \det H_{\theta} - \log Z(\theta)$$
$$\text{where } \widehat{f}_{\theta}^{(i)} = \underset{f}{\operatorname{argmin}}\, \ell \left(D_{\text{train}}^{(i)}, f \right) + \lambda R_{\theta} \left(f \right)$$

This is bilever optimization. Even though it lost the convexity, it is tractable and can be solved in practice by alternating gradient descent. I claim that gradient descent of the outer problem = training a transformer, gradient descent of the inner problem is a transformer forward pass.

7 Inner problem

For a dataset $D = \{(x_i, y_i)\}$ we have

$$\widehat{f}_{\theta} = \underset{f}{\operatorname{argmin}}\, \ell \left(D, f \right) + \lambda R_{\theta} \left(f \right) = \underset{f}{\operatorname{argmin}}\, \| f(x_i) - y_i \|_2^2 + \lambda \langle \theta, \Phi(f) \rangle$$

Note, that if Φ is convex in f then this can be solved by subgradient descent on f:

$$f^{(t+1)} = f^{(t)} - \eta \left[2 \sum_{i} (f^{(t)}(x_i) - y_i) \delta_{x_i} + \lambda\, g^{(t)} \right]$$

where $g^{(t)} \in \partial_f R_{\theta}(f^{(t)})$ is a subgradient of the regularizer.

7.1 How transformer forward pass can implement a (sub)gradient descent

Now lets go through some of the examples in Section 3.3 to see how transformer-like networks can implement a (sub)gradient g descent for the inner problem.

Energy regularization

$$\Phi : L^2(M) \to \mathbb{R}, \qquad \Phi(f) := \|f\|_{L^2(M)}^2 \qquad R_{\theta}(f) = \theta \|f\|_{L^2(M)}^2 \qquad g = 2f$$

Anisotropic Sobolev Regularization Parameter θ is a measure over PSD matrices.

$$\Phi(f)(Q) := \|\nabla f(x)\|_Q^2 \qquad R_{\theta}(f) = \int_{Q \in \mathbf{S}_+} \|\nabla f(x)\|_Q^2 d\theta(Q) \qquad g = -\int_{Q \in \mathbf{S}_+} \Delta_Q f d\theta(Q)$$

Anisotropic second-order TV

$$\Phi : X \to \mathcal{C}(\mathcal{S}_+, \mathbb{R}_+), \qquad \Phi(f)(Q) := \|\Delta_Q f\|_{\mathcal{M}}, \qquad g = \int_{\mathcal{S}_+} \Delta_Q \operatorname{sign}(\Delta_Q f) d\theta(Q)$$

where $\Delta_Q f := \nabla \cdot (Q \nabla f)$.

As the functions are defined on a "flat" domain $\mathcal{M} \subset R^d$, the discrete version of Δ_Q looks like:

$$L \in R^{n \times n} : L_{ij} = \exp\left(\frac{-\|x_i - x_j\|_{Q^{-1}}^2}{\epsilon} \right)$$

If our domain is curved manifold $\mathcal{M}$ we can W.L.O.G. (by Takahashi's thm hehe) assume that it's $\mathcal{M} \in S^{d-1}$, hence, the discrete version of Δ_Q will require projection $P_i = I - x_i x_i^{\top}$:

$$\begin{aligned} L \in R^{n \times n} : L_{ij} &= \exp\left(\frac{-(x_i - x_j)^{\top} P_i Q^{-1} P_i (x_i - x_j)^{\top}}{\epsilon} \right) \\ &= \exp\left(-\frac{\left(x_j - \left(x_i^{\top} x_j\right) x_i\right)^{\top} Q^{-1} \left(x_j - \left(x_i^{\top} x_j\right) x_i\right)}{\epsilon} \right) \end{aligned}$$

Note, that all the exp terms are kind of similar to the original transformer softmax terms. I think, softmax should be replaced by those. The final idea is that a transformish NN with infinite amount of heads approximates the disctribution over Q.

Theorem 1 (Stability from coercive Φ-regularization). *Let $(X, \|\cdot\|_X)$ be a reflexive Banach space of functions on a domain M, and let*

$$R_\theta(f) := \langle \theta, \Phi(f) \rangle \in [0, \infty]$$

where $\Phi : X \to V$ is a measurable map into a real locally convex space V and $\theta \in V^*$ is such that R_θ is a proper, convex, lower semicontinuous functional on X. Assume the following:

1. ***Coercivity (modulo a nullspace).*** *There exists a finite-dimensional subspace $\mathcal{N} \subset X$ and a constant $c > 0$ such that for all $f \in X$,*
$$R_\theta(f) \;\geq\; c\,\|f - \Pi_{\mathcal{N}} f\|_X - c,$$
where $\Pi_{\mathcal{N}}$ denotes a fixed linear projection onto $\mathcal{N}$. In particular, the sublevel sets $\{f : R_\theta(f) \leq C\}$ are relatively compact in $X/\mathcal{N}$.

2. ***Strong convexity of the empirical loss.*** *For each dataset $D = \{(x_i, y_i)\}_{i=1}^n$, define*
$$\mathcal{L}_D(f) := \frac{1}{n} \sum_{i=1}^n \ell\big(f(x_i), y_i\big),$$
and assume $\mathcal{L}_D$ is α-strongly convex on $X/\mathcal{N}$ with respect to $\|\cdot\|_X$, uniformly over all datasets D (for some $\alpha > 0$).

3. ***Lipschitz dependence on function values.*** *Assume $\ell(\cdot, y)$ is L-Lipschitz in its first argument for all y, and that point-evaluation is continuous on X in the sense that there exists $\kappa > 0$ such that for all $x \in M$ and all $f, g \in X$,*
$$|f(x) - g(x)| \leq \kappa \|f - g\|_X.$$

Observed data $y_i = f^*(x_i) + \mathcal{N}(0, \sigma^2)$

$$\widehat{f} = \underset{f}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^n \ell(f(x_i), y_i) + \lambda R(f) \Rightarrow \|f^* - \widehat{f}\|_X \leq C(n, \sigma, R, \lambda)$$

Proposition 1 (Bilevel learning of θ as a MAP surrogate). *Let $R_\theta(f) = \langle \theta, \Phi(f) \rangle$ and define the inner estimator*

$$\widehat{f}_\theta(D) \in \arg\min_{f \in X} \Big\{ \mathcal{L}_D(f) + \lambda R_\theta(f) \Big\}.$$

Consider the bilevel objective based on masking (or train/validation splits),

$$\min_\theta \; \mathbb{E}_{mask}\Big[\mathcal{L}_{\mathrm{val}}\big(\widehat{f}_\theta(D_{\mathrm{train}})\big) \Big] + \Omega(\theta).$$

Then bilevel optimization selects θ so as to minimize the predictive risk of the MAP estimator induced by the Gibbs prior

$$p_\theta(df) \propto \exp\big(-\lambda R_\theta(f)\big)\, \nu(df).$$

In contrast, maximum-likelihood learning of θ for the induced Gibbs model includes an additional global complexity term $\log Z(\theta)$*, which is generally not captured by the bilevel objective unless explicitly approximated (e.g., via Laplace/Occam corrections).*